CS316

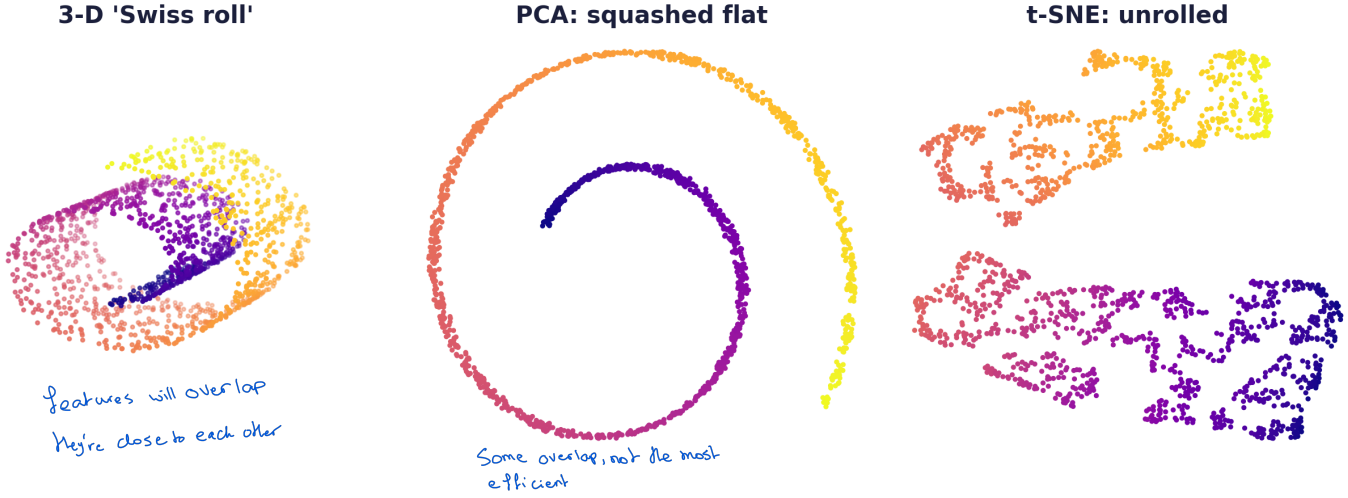
INTRODUCTION TO AI AND DATA SCIENCE

Topic 7.3

Dimensionality reduction for visualizing complex data
Introduction to t-SNE

Instructor: Dr. Heyfa Ammar

PCA is linear. Real data often is not



PCA keeps global, straight-line variance. It can crush curved structure. **t-SNE** keeps local neighborhoods, so curved structure survives.

t-SNE

t-SNE (t-Distributed Stochastic Neighbor Embedding) is a dimensionality reduction technique primarily used for visualizing high-dimensional data in lower-dimensional spaces (usually 2D or 3D). It is particularly useful when you have complex, high-dimensional (many columns) data and want to visualize patterns, clusters, or relationships within that data.

Here's a quick breakdown of how it works:

1. **High-Dimensional Space:** t-SNE starts with data points in a high-dimensional space, where distances between data points are meaningful.
2. **Pairwise Similarities:** t-SNE calculates the pairwise similarities between data points. In the high-dimensional space, it measures how likely two points are to be neighbors based on their distances.

t-SNE (contd)

3. Low-Dimensional Mapping: It then projects the data into a lower-dimensional space (2D or 3D). The goal is to minimize the difference between the pairwise similarities in the high-dimensional and low-dimensional spaces.

4. Preserving Local Structure: The main feature of t-SNE is that it preserves local structure—meaning points that are close to each other in the high-dimensional space will also be close to each other in the lower-dimensional projection. However, t-SNE does not preserve global distances or the exact shape of the data, so it's primarily used for visualization rather than interpretation of precise relationships.

5. Perplexity: One key hyperparameter in t-SNE is "perplexity," which balances the attention between local and global aspects of the data. Larger values focus more on global structure, and smaller values focus on local clusters. Max distance to consider 2 points are neighbors/same cluster

t-SNE (contd)

Pros:
Excellent for visualizing high-dimensional data.

Can uncover structure or clusters that aren't easily seen in raw data.

Cons:

Computationally expensive, especially with large datasets.

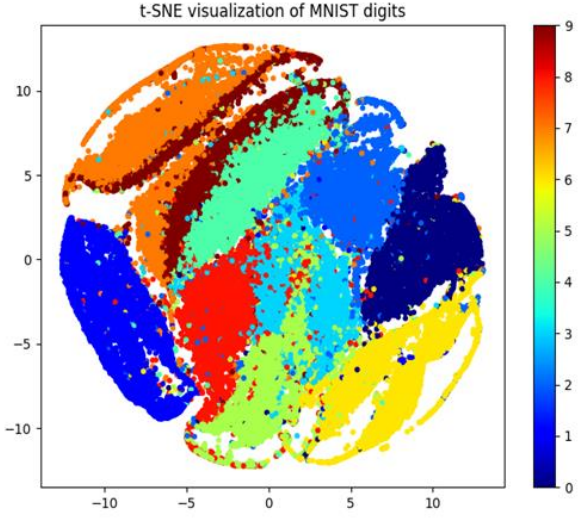
It's more of a tool for visualization and not for preserving exact relationships in the data.

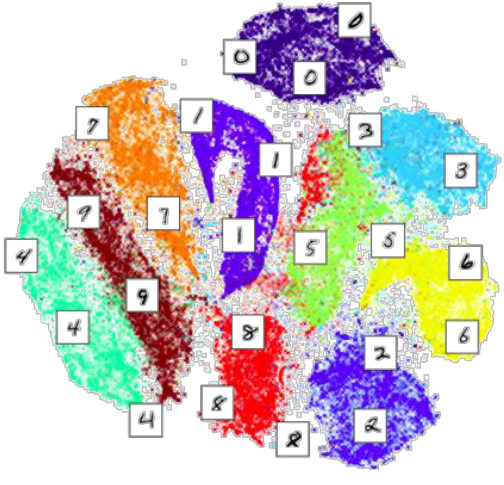
PCA makes a new dataset with less feature (transforms new data into the new space)

t-SNE can be used after PCA

Example: MNIST dataset

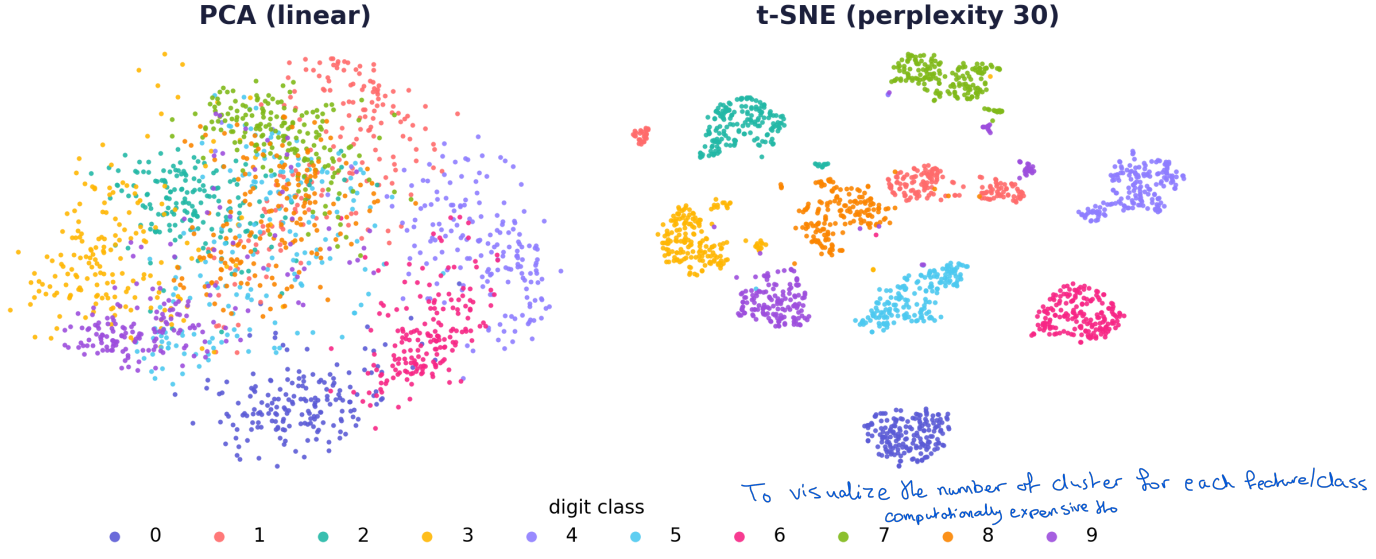






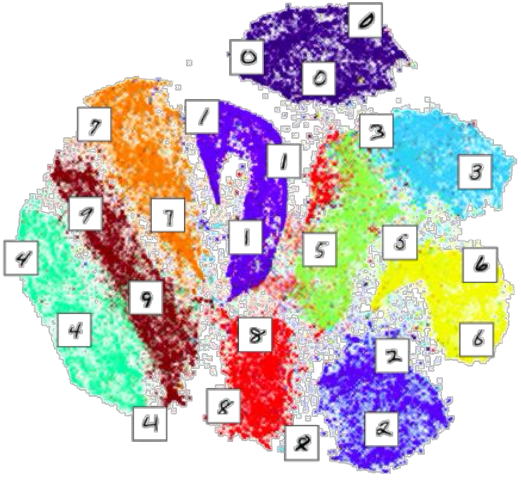
grp each digit as color, some digit could be in incorrect classes

The payoff: PCA vs t-SNE on real digits



Colors are used only for display. Neither method saw the labels.

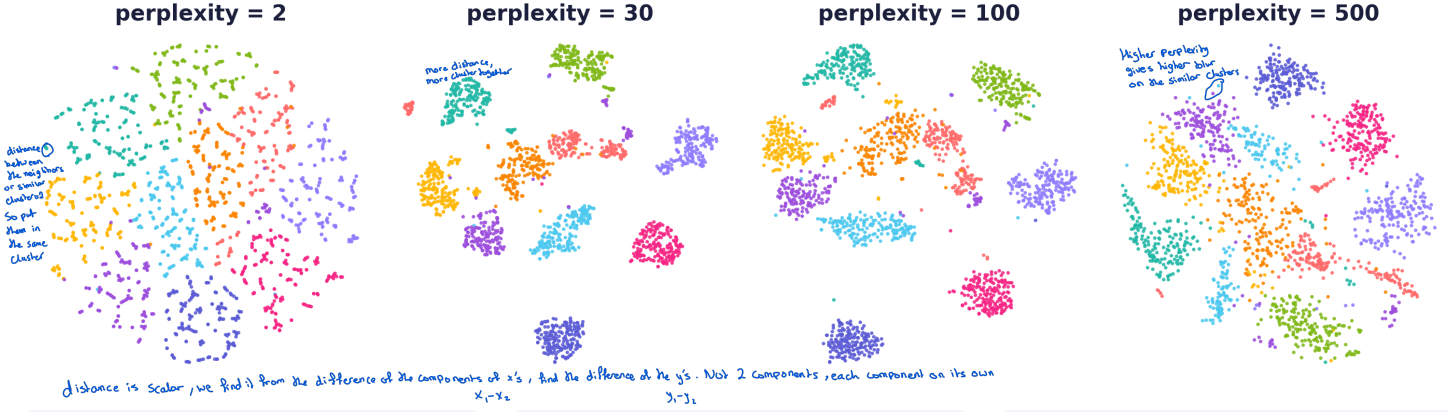
Examples



Interpretation:

1. We see clear clusters of our data.
2. We see some overlap between similar digits (5, 8, and 3)
3. t-SNE is very expensive to run, pair-wise similarity is harder to extract than variance (with PCA).

The main knob: perplexity



Too low

Tiny fragments, noisy confetti

Sweet spot

Clear local groups, typically 5 to 50

Too high

Clusters blur and merge

Reading a t-SNE plot: trust vs. don't trust

Imp to know how to read

OK You can trust

- Points close together are similar in the original space.
- Tight groups usually indicate real local structure. Condensed points
- Outliers and mislabeled items often sit in the wrong island. Some points are isolated

NO Don't over-read

- Distance between clusters means little. not the real distance in the original space
- Cluster size and density are not reliable.
- Axes have no meaning; rerunning can rotate or flip the map. transforming our space to smaller dimension wouldn't give the same shape
- Random data can still show fake-looking clusters.

Choosing a method: PCA vs t-SNE

	PCA	t-SNE
Type	Linear	Non-linear
Preserves	Global variance	Local neighborhoods
Speed on big data	Very fast	Slow because of the calculations of the distance
Transforms new points	Yes	No (refit needed)
Best used for	Preprocessing, features	Visual exploration not used for feature engineering
	PCA	t-SNE

Your turn: think, pair, share (10 min)

Scenario 1

You ran t-SNE once and saw 5 clusters in your customer data. What would you do before reporting it?

we should change the perplexity
run different tests with different random seeds/ States

Scenario 2

A teammate says 'clusters A and B are closer than A and C, so A is more similar to B.' Is that safe to claim?

No bcuz distance in the plot have no meaning

Scenario 3

You want a 2-D input for a classifier trained today and used on new data tomorrow. PCA or t-SNE? Why?

Quick check: true or false?

1. The x and y axes of a t-SNE plot have a fixed meaning.
2. Perplexity roughly sets how many neighbors each point 'pays attention to'.
3. A bigger cluster on the plot always means a bigger group in the data.
4. You can apply a trained t-SNE model directly to new data points.
5. Running PCA before t-SNE can speed it up and reduce noise.

Key takeaways

1. t-SNE maps high-D data to 2D while keeping neighborhoods together.
2. It works in three steps: measure neighbors, place points (adjust by minimizing KL divergence).
3. Perplexity is the main knob. Try several values and several seeds.
4. Read it as a map of neighborhoods. Do not trust distances, sizes, or axes.

Homework: run t-SNE on a dataset of your choice and compare two perplexity values.

References and further reading

- **van der Maaten, L. & Hinton, G. (2008).** Visualizing Data using t-SNE. Journal of Machine Learning Research, 9, 2579-2605.
- **Wattenberg, M., Viégas, F. & Johnson, I. (2016).** How to Use t-SNE Effectively. Distill. (interactive demos of every pitfall)
- **scikit-learn documentation.** sklearn.manifold.TSNE and the Manifold learning user guide.
- **McInnes, L., Healy, J. & Melville, J. (2018).** UMAP: Uniform Manifold Approximation and Projection. arXiv:1802.03426.